

IE 203 - Operations Research II

Midterm

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Duration: 1.5 hours

Important notice: 1. This is a CLOSED BOOK exam.

2. You may keep the questions.

Question I (30 pts.)

Consider the mixed integer nonlinear programming (MINLP) problem

$$\begin{array}{ll} \max & z = -4x_1 - x_2 \\ \text{s.t.} & 2x_1 + 3x_2 \geq 4 \\ & 2x_1^2 + x_2 \geq 2 \\ & -x_1 + x_2 \leq 3 \\ & x_1, x_2 \geq 0 \\ & x_2 \text{ integer} \end{array}$$

- Show graphically the feasible solution set.
- Solve graphically the MINLP.
- Show graphically the convex hull of the feasible solution set.
- Find the extreme points and extreme directions of the convex hull of the feasible solution set.
- Solve graphically the LP consisting of the minimization of the objective function subject to the constraints describing the convex hull. Is the optimal solution same as the one of the original MINLP obtained in (b)?

Question II (40 pts.)

Consider the mixed integer linear programming (MILP) problem

$$\begin{array}{ll} \text{MILP: } z_{IP} = \max & 8x_1 + 5x_2 \\ \text{s.t.} & x_1 + x_2 \leq 6 \\ & 9x_1 + 5x_2 \leq 45 \\ & x_1, x_2 \geq 0 \\ & x_2 \text{ integer.} \end{array}$$

- Solve the LP relaxation of the problem using simplex algorithm and obtain optimal tableau.
- Generate the Gomory cut corresponding to the basic variable violating integrality restriction.
- Add the new cut to the obtained optimal tableau and reoptimize it using dual simplex algorithm for finding the next LP solution.
- Show graphically feasible solution set with the generated Gomory cut.

Question III (30 pts.)

Consider the following unconstrained problem:

$$\min f(x_1, x_2) = x_1^2 - x_1x_2 + 2x_2^2 - 2x_1 + e^{x_1+x_2}$$

- Write the first-order necessary optimality conditions. Are these conditions also sufficient for optimality? Why?
- Is $\mathbf{x}^* = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ an optimal solution? If not, identify the steepest descent direction at $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and the step length using exact line search along the steepest descent direction.
- Repeat (b) for Newton's method. You do not have to calculate the step length. It is enough to give the function in α , which is to be minimized.
- Solve this problem after dropping the last term in the objective function.